Smart Health 36 (2025) 100560

Contents lists available at ScienceDirect

Smart Health

journal homepage: www.elsevier.com/locate/smhl

SNOMED CT ontology multi-relation classification by using knowledge embedding in neural network

Bofan He *, Jerry Q. Cheng, Huanying Gu

Department of Computer Science, New York Institute of Technology, USA

ARTICLE INFO

Keywords:
Link prediction
SNOMED CT
Knowledge graph embeddings
TransE
Multi-head classifier
Ontology enrichment
Healthcare informatics

ABSTRACT

SNOMED CT is a widely recognized healthcare terminology designed to comprehensively represent clinical knowledge. Identifying missing or incorrect relationships between medical concepts is crucial for enhancing the scope and quality of this ontology, thereby improving healthcare analytics and decision support. In this study, we propose a novel multi-link prediction approach that utilizes knowledge graph embeddings and neural networks to infer missing relationships within the SNOMED CT knowledge graph. By utilizing TransE, we train embeddings for triples (concept, relation, concept) and develop a multi-head classifier to predict relationship types based solely on concept pairs. With an embedding dimension of 200, a batch size of 128, and 10 epochs, we achieved the highest test accuracy of 91.96% in relationships prediction tasks. This study demonstrates an optimal balance between efficiency, generalization, and representational capacity. By expanding on existing methodologies, this work offers insights into practical applications for ontology enrichment and contributes to the ongoing advancement of predictive models in healthcare informatics. Furthermore, it highlights the potential scalability of the approach, providing a framework that can be extended to other knowledge graphs and domains.

1. Introduction

Knowledge graphs have emerged as a foundational tool for representing complex relationships between entities across various domains, particularly in healthcare. Medical ontologies are structured vocabulary systems designed to standardize and precisely express medical concepts and their interrelationships, thereby enhancing communication and information exchange among healthcare professionals while reducing misunderstandings. SNOMED CT (Systematized Nomenclature of Medicine – Clinical Terms) (El-Sappagh, Franda, Ali, & Kwak, 2018), one of the most extensively used medical ontologies, provides a comprehensive and hierarchical collection of terms spanning various medical fields. As a unified and precise terminology system, SNOMED CT has gained wide-ranging global adoption and recognition within the international medical and health communities. It plays an essential role in facilitating the accurate representation and understanding of medical concepts, contributing to improved clinical decision-making, research, and interoperability across healthcare systems.

Despite its comprehensiveness, the incomplete nature of relationships within SNOMED CT limits its potential for broader application. To address this gap, multi-link prediction aims to identify new links and their corresponding types, thereby enriching the ontology's relational structure.

This paper introduces a novel multi-link prediction framework that leverages knowledge graph embeddings. Through the application of TransE, embeddings are generated for concept–relation–concept triples, and a multi-head classifier is implemented to

* Corresponding author.
E-mail addresses: bhe@nyit.edu (B. He), jcheng18@nyit.edu (J.Q. Cheng), hgu03@nyit.edu (H. Gu).

https://doi.org/10.1016/j.smhl.2025.100560
Received 10 March 2025; Accepted 15 March 2025
Available online 26 March 2025
2352-6483/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

predict relationship types between concept pairs. By eliminating the explicit requirement for relation embeddings during inference, this approach offers a scalable and flexible solution for predicting missing links within the knowledge graph. Moreover, this study explores the potential to generalize the methodology across diverse datasets, thereby emphasizing its robustness and versatility. The findings from this study extend beyond SNOMED CT, offering methodologies that are applicable to various domain-specific knowledge graphs.

2. Background and related work

The September 2024 US Edition of SNOMED CT incorporates data from the July 2024 International Edition and the September 2024 US Extension. It comprises 375,656 concepts, 1,363,082 descriptions, and 1,304,539 relationships. Also, in the September 2024 US Extension, 104 new active concepts, 295 new active descriptions, and 1121 new active relationships were introduced, while 103 concepts were newly inactivated. The extension also reports 3838 sufficiently defined concepts and 4627 primitive concepts. These enhancements reflect SNOMED CT's ongoing evolution to support comprehensive medical knowledge representation and its adaptability to meet evolving healthcare needs.

The VERSE framework proposed by Tsitsulin, Mottin, Karras, and Müller (2018) introduces a versatile, scalable, and task-adaptive approach to graph embeddings, explicitly designed to capture vertex-to-vertex similarity measures. Unlike earlier methods that either relied on rigid local neighborhood definitions (e.g., DeepWalk (Perozzi, Al-Rfou, & Skiena, 2014), Node2Vec (Grover & Leskovec, 2016)) or computationally expensive matrix factorization (e.g., HOPE (Ou, Cui, Pei, Zhang, & Zhu, 2016), LINE (Tang et al., 2015)), VERSE integrates flexibility and efficiency by minimizing the Kullback–Leibler divergence between real and embedded similarity distributions. By combining elements of traditional embedding methods with neural network architectures, this framework bridges the gap between computational efficiency and relational expressiveness, making it a robust solution for knowledge graph enrichment. Wu, Song, Ge, and Ge (2022) evaluate Graph Neural Networks (GNNs) for link prediction, addressing key challenges in current practices such as underreported performance, inconsistent evaluation settings, and unrealistic negative sampling strategies. They propose a new benchmarking framework, including the Heuristic Related Sampling Technique (HeaRT), to generate hard negative samples that better reflect real-world scenarios. Their findings highlight the need for robust evaluation metrics and improved methodologies, which are particularly relevant for domains like SNOMED CT, where distinguishing complex relationships in medical ontologies is crucial. Biswas (2020) present an embedding-based approach for link prediction in knowledge graph completion, addressing challenges of sparsity and missing links in knowledge graphs. The proposed model integrates structural information and textual entity descriptions using an attention-based encoder–decoder framework, achieving significant improvements in head and tail prediction as well as triple classification compared to state-of-the-art methods.

However, the aforementioned methods and frameworks face significant limitations when applied to SNOMED CT due to the extensive size of its knowledge graph and the large number of unique concepts. The sheer scale of SNOMED CT, with hundreds of thousands of concepts and relationships, poses scalability challenges that these existing methods may struggle to address effectively. This highlights the necessity for more robust, scalable, and specialized methodologies tailored to handle the unique complexities of SNOMED CT and similar large-scale medical ontologies.

Existing methodologies for link prediction in medical ontologies span statistical models, embedding-based frameworks such as TransE (Bordes, Usunier, Garcia-Duran, Weston, & Yakhnenko, 2013), and graph neural networks (GNNs). Embedding-based methods like TransE excel in modeling triples by optimizing low-dimensional representations, while GNNs provide relational inductive biases, making them suitable for reasoning tasks. However, embedding methods generally offer simpler architectures and greater computational efficiency. This work capitalizes on TransE's simplicity and integrates it with a neural network-based multi-head classifier for accurate relation type prediction. Moreover, it addresses the limitations of scalability and computational overhead, often associated with GNNs, presenting a hybrid approach that balances efficiency and predictive accuracy. By integrating elements of traditional embedding methods with neural network architectures, this framework bridges the gap between computational efficiency and relational expressiveness, making it a robust solution for knowledge graph enrichment.

A study by Abeysinghe et al. (2023) introduced a deep learning approach to detect missing is-a relationships in SNOMED CT. Their method focused on concept pairs where one concept's name is a proper subset of another, using this pattern to predict overlooked hierarchical links. By combining concept name embeddings, hierarchical features derived from Graph Convolutional Networks (GCNs), enriched lexical attributes, and logical definition embeddings, their binary classification model achieved a high precision of 83.48% in identifying missing is-a relations. Their findings highlight the potential of deep learning techniques in improving ontology quality assurance, outperforming traditional rule-based approaches. Similarly, our previous research (He, Cheng, & Gu, 2025) investigated the use of static and dynamic embedding methods to classify is-a relationships in SNOMED CT. The study applied both word embedding-based neural networks and transformer-based models such as BERT and RoBERTa, and compared their effectiveness with traditional classifiers like SVM, KNN, and MLP. The results showed that RoBERTa outperformed these conventional methods, achieving a precision of 0.933. This underscores the advantages of transformer-based models in refining hierarchical structure identification within SNOMED CT, ultimately reducing inconsistencies and improving the ontology's reliability.

Building on these studies, our research moves beyond binary is-a classification to a broader multi-relation classification task within SNOMED CT. Rather than focusing solely on containment-based patterns, we employ knowledge embedding techniques to capture the semantic structure of multiple relationship types.

3. Method

Our methodology comprises two primary components:

Embedding Generation with TransE: TransE is employed to generate low-dimensional embeddings for entities and relations. The scoring function ensures that the embeddings of a triple (concept A, relation, concept B) are consistent with the relation, where the embeddings of the source concept (concept A), the relation, and the destination concept (concept B) are represented, respectively. The design of this embedding mechanism ensures that the learned representations effectively capture the semantic relationships between entities and their relational contexts. This embedding process not only encodes the structural information of the knowledge graph but also provides a flexible basis for downstream prediction tasks.

Work Principle: TransE is an embedding model for multi-relational knowledge graphs that represents entities and relationships in a low-dimensional vector space. The fundamental idea is to model relationships as translations in the embedding space. For a valid triplet (h, r, t), TransE enforces the relation:

$$\mathbf{h} + \mathbf{r} \approx \mathbf{t}$$

where $\mathbf{h}$, $\mathbf{r}$, and $\mathbf{t}$ are the embeddings of the head, relation, and tail, respectively. The model assumes that the embedding of the tail $\mathbf{t}$ should be close to the embedding of the head $\mathbf{h}$ translated by the relation $\mathbf{r}$.

Objective Formula: The learning objective in TransE minimizes a margin-based ranking loss to differentiate valid triplets from corrupted ones:

$$L = \sum_{(h,r,t)\in S} \sum_{(h',r,t')\in S'} \left[\gamma + d(\mathbf{h} + \mathbf{r}, \mathbf{t}) - d(\mathbf{h}' + \mathbf{r}, \mathbf{t}')\right]_+$$

where: S: Set of valid triplets in the training data. S': Set of corrupted triplets created by replacing h or t in S with random entities. $\gamma > 0$: Margin hyperparameter. $d(x, y)$: Dissimilarity measure, often L_1 or L_2-norm. $[x]_+ = \max(0, x)$: Ensures non-negative loss. This loss encourages valid triplets to have lower dissimilarity scores than corrupted triplets.

Motivation for Adaptation: In this study, the TransE framework is adapted to focus on *relation prediction tasks*, where the goal is to predict the missing relation r given a pair of entities h and t. To align the model with this objective, the fundamental principle is modified from:

$$\mathbf{h} + \mathbf{r} \approx \mathbf{t} \quad to: \quad \mathbf{h} - \mathbf{t} \approx \mathbf{r}$$

This adjustment shifts the focus from entity prediction to relation prediction by ensuring that the difference between the embeddings of the head and tail entities approximates the embedding of the relation. This formulation better captures the nuances of predicting relationships in a knowledge graph, especially in tasks where inferring the nature of the relationship is crucial. The adapted loss function reflects this modification by optimizing the embeddings accordingly.

Multi-Head Classifier: A multi-head neural network classifier is built on top of the embeddings. Given the embeddings of a pair of concepts (concept A, concept B), the classifier predicts the most probable relationship type. This design eliminates the need for explicit relation embeddings during inference. By relying on the computed difference vector, the classifier achieves enhanced scalability and computational simplicity. This approach further streamlines the inference process, enabling an efficient integration into real-world applications.

4. Experimental setup

Data Preparation: The dataset utilized in this study originates from the 2024 release of the SNOMED CT United States (US) Edition. The data package includes a relation table that specifies the source concept ID, relation type ID, and destination concept ID. Additionally, it contains a description table that provides textual mappings for each unique identifier, facilitating the interpretation of concepts and relations within the dataset. The relation table was employed to extract structured triples, while the description table was used to map IDs to human-readable descriptions.

As part of the preprocessing step, a new structured dataset was created, comprising over 463,000 unique entities and 133 relation types as source concepts, relation types, and destination concepts as input triples in the format (head, relation, tail). This structured dataset serves as the foundation for both training and evaluation. Entities, including head and tail concepts, as well as relations, were mapped to unique numeric IDs. The dataset was then converted into numeric triples in the form of [head_id, relation_id, tail_id], which ensures compatibility with the embedding and classification models. To ensure balanced and representative data for model training and evaluation, the dataset was divided into training (80%), validation (10%), and test (10%) subsets. During this process, only active concepts were retained, further enhancing the reliability and relevance of the evaluation framework.

Neural Network Architecture: The architecture comprises two primary components:

Embedding Layer (TransE): This layer is responsible for learning vector representations for entities and relations within the knowledge graph. The embeddings are initialized using the Xavier uniform distribution, which ensures stability during training by maintaining appropriate weight scales. During both initialization and the forward pass, embeddings are normalized to maintain consistent vector magnitudes, which is essential for the stability and performance of distance-based models like TransE. The output of this layer comprises normalized embeddings for both head and tail entities, forming the basis for subsequent relational reasoning.

Relation Classifier: The relation classifier predicts the relationship type between a given pair of entities, represented by their embeddings for both head and tail. The input to this component consists of the entity embeddings for the head and tail. Relation

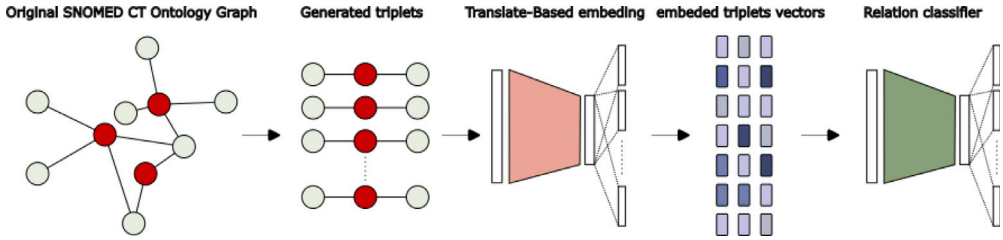

Fig. 1. Pipeline for translation-based graph embedding and relation classifier method.

classification is modeled by predicting the difference vector between the tail and head embeddings, aligning with the relational reasoning captured by the TransE framework. The classifier incorporates a linear layer to map the difference vector to logits corresponding to each possible relation type. Dropout regularization is applied at a rate of 30% to mitigate overfitting, while normalization ensures stable input distributions to the linear layer. This design framework enhances the model's generalizability, enabling robust performance on unseen data.

Fig. 1 shows a pipeline for translation-based graph embedding and relation classification beginning with the SNOMED CT ontology graph, where nodes represent medical concepts, and edges represent relationships. This graph is decomposed into triplets in the form (head, relation, tail), which are then processed using the translation-based embedding model (TransE) to map entities and relations into a continuous vector space. The resulting embedded vectors capture semantic and structural information from the graph. These vectors are then input into a relation classifier, which predicts or classifies the relationships between entities. This approach facilitates the analysis and inference of knowledge in the ontology.

Training and Evaluation:The training and evaluation process involves three key stages: training, validation, and testing. During the training phase, the embedding and classifier components are jointly trained using the Adam optimizer with a learning rate of 1e−3. Cross-entropy loss is used as the loss function to optimize the model, ensuring the accurate prediction of relations. Validation occurs at the end of each epoch, where the model's performance is assessed using validation loss and accuracy metrics. The model with the highest validation accuracy is saved as the best model. Finally, the testing phase evaluates the best-performing model on the test set to compute the test loss and accuracy, providing a comprehensive assessment of the model's generalization capabilities. This workflow is designed to rigorously evaluate the model's predictive performance while ensuring robust and reliable benchmarks for iterative improvement.

*Experimental Hyperparameters Configuration:*Embedding dimension: 200, Batch size: 128, Epochs: 10, Device: GPU. Metrics: Training, validation, and test metrics include loss, accuracy metrics.

Evaluation Metrics: The experiments varied three critical hyperparameters: embedding dimensions, batch sizes, and the number of training epochs, as these parameters significantly influence model performance by balancing representation capacity, training stability, and computational efficiency. The learning rate was fixed at 1e−3 across all runs to maintain consistency. Training was conducted on an NVIDIA A100 GPU to leverage its computational capabilities for efficient model optimization. Metrics such as training loss, validation loss, training accuracy, and validation accuracy were recorded at each epoch to monitor and evaluate model performance.

5. Results

Table 1 includes key parameters such as embedding dimensions, batch sizes, and the number of epochs, along with the final training, validation, and test accuracies. These parameters were selected based on their critical impact on the model's ability to generalize and perform efficiently. Embedding dimensions were chosen to balance representational capacity and computational complexity, while batch sizes and epochs were varied to explore their effects on model stability and convergence. The inclusion of test accuracy allows for a direct comparison of the generalization performance of each model configuration.

The highest test accuracy of 91.96% was achieved in Run 3, which used an embedding dimension of 200, a batch size of 128, and 10 epochs. This result highlights the effectiveness of a moderate embedding dimension and batch size combination in providing sufficient representational capacity and efficient training.

In terms of embedding dimensions, the experiments showed that an embedding size of 200 consistently produced strong results across multiple runs, with test accuracies exceeding 91.5% in Runs 2, 3, 7, and 8. Increasing the embedding dimension to 400 in Run 6 did not significantly improve test accuracy, suggesting diminishing returns for larger embedding sizes. Batch size also played a significant role in model performance. Batch sizes of 128 produced the most stable results, as seen in Runs 3, 4, 7, 10, and 11, where test accuracies ranged from 91.06% to 91.96%. Smaller batch sizes, such as 64 in Runs 2, 8, and 9, also performed well, but training times were longer, and test accuracies were slightly lower compared to batch size 128. Larger batch sizes of 256 in Runs 5 and 6 showed stable performance, with test accuracies of 91.46% and 91.65%, respectively, but they did not outperform the configurations with batch size 128.

The number of training epochs showed that 10 epochs were sufficient for this task, as evidenced by the consistent performance across most runs. Extending training to 20 epochs in Run 1 did not yield significant gains, with a test accuracy of 91.64%, suggesting that the model had already converged within the first 10 epochs.

Table 1
Experimental results.

Run	Embedding dim	Batch size	Epochs	Train accuracy	Validation accuracy	Test accuracy
1	100	32	20	95.20%	89.92%	91.64%
2	200	64	10	95.25%	90.73%	91.81%
3	200	128	10	95.41%	90.74%	91.96%
4	300	128	10	95.39%	90.73%	91.71%
5	200	256	10	95.43%	90.42%	91.46%
6	400	256	10	95.45%	90.48%	91.65%
7	200	128	10	95.29%	90.58%	91.64%
8	200	64	10	95.17%	90.73%	91.79%
9	200	64	10	95.27%	90.67%	91.19%
10	200	128	10	95.37%	90.56%	91.06%
11	200	128	10	95.38%	90.50%	91.10%

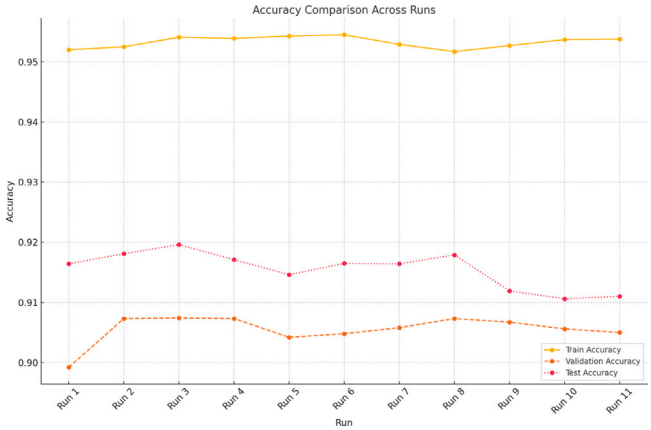

Fig. 2. Accuracy comparison across runs (each line represents one type of accuracy, allowing for a clear visual comparison of performance across different configurations).

Validation accuracies across all runs ranged from 89.92% to 90.74%, with the highest validation accuracy observed in Run 3. These observations align with the overall conclusions about model performance, highlighting its consistency and robustness in distinguishing complex relationships within the dataset. These results were consistent with test accuracies, indicating good generalization. Training accuracies were consistently high, ranging from 95.17% to 95.45%, across all runs. The small gap between training and validation accuracies demonstrated the stability of the models and minimal overfitting.

Fig. 2 illustrates the progression of training, validation, and test accuracies across all experimental runs. Each line represents a different run, showcasing the performance trends under varying configurations of embedding dimensions, batch sizes, and training epochs. The chart enables a comparative view of how each model generalizes to unseen data (validation and test sets) while minimizing overfitting (measured by the gap between training and validation accuracies).

From the chart, it is evident that training accuracies are consistently high across all runs, with final values ranging between 95.17% and 95.45%. This indicates that the models effectively learned the relationships in the training data. Validation accuracies display less variability, ranging from 89.92% to 90.74%, reflecting stable generalization capabilities across different hyperparameter settings. Test accuracies, measured on unseen data, closely align with validation accuracies, suggesting that the models are not overfitting and generalize well to new data.

Runs with an embedding dimension of 200 and batch sizes of 128 (e.g., Runs 3 and 7) achieved the highest test accuracies, with Run 3 being the best performer at 91.96%. In contrast, configurations with smaller embedding dimensions, such as 100 in Run 1, resulted in slightly lower test accuracy at 91.64%, possibly due to insufficient representational capacity. Larger embedding dimensions (e.g., 400 in Run 6) did not significantly outperform moderate dimensions, indicating that increasing embedding size beyond 200 offers diminishing returns.

Batch size also influenced the results. Runs with batch sizes of 128 consistently performed well, while smaller batch sizes (e.g., 64) showed comparable but slightly lower test accuracies, likely due to longer convergence times. Larger batch sizes (e.g., 256) resulted in stable yet slightly lower performance compared to batch size 128.

The small and consistent gaps between training, validation, and test accuracies across all runs indicate the robustness of the models and minimal overfitting. This is particularly evident in top-performing configurations, such as Run 3, where training, validation, and test accuracies align closely, demonstrating strong generalization.

6. Discussion

The results of this study validate the potential of integrating TransE embeddings with a multi-head classifier for link prediction tasks. The stabilization of validation accuracy at higher epochs suggests robust generalization. Future enhancements could include adaptive optimization strategies or more complex architectures to further improve performance.

While showing promise, the model's performance is subject to several limitations. First, evaluation was conducted on a limited set of relation types, which raises concerns about the model's ability to generalize to unseen relations. Second, the use of fixed embedding dimensions may hinder the effective representation of complex and nuanced relationships. Finally, the scalability of the current framework remains untested, particularly concerning its ability to handle larger datasets or datasets with significantly greater relational diversity.

Future work should focus on addressing the identified limitations. This includes developing dynamic embedding techniques to better capture the nuances of relationships, testing the model's generalizability on cross-domain datasets, and exploring the integration of multimodal data, such as text and images, to enrich relational reasoning. Additionally, investigating hybrid models that combine Graph Neural Networks (GNNs) with TransE embeddings is crucial for enhancing both relational reasoning capabilities and scalability.

CRediT authorship contribution statement

Bofan He: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jerry Q. Cheng:** Conceptualization, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing. **Huanying Gu:** Conceptualization, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was partially supported by the National Science Foundation, United States Grants CNS-2120350, III-2311598.

Data availability

The authors do not have permission to share data.

References

Abeysinghe, Rashmie, Zheng, Fengbo, Bernstam, Elmer V., Shi, Jay, Bodenreider, Olivier, & Cui, Licong (2023). A deep learning approach to identify missing is-a relations in SNOMED CT. *Journal of the American Medical Informatics Association: JAMIA*, *30*(3), 475–484. http://dx.doi.org/10.1093/jamia/ocac248.

Biswas, R. (2020). Embedding based link prediction for knowledge graph completion. In *Proceedings of the 29th ACM international conference on information & knowledge management* (pp. 3221–3224). New York, NY, USA: ACM, http://dx.doi.org/10.1145/3340531.3418512.

Bordes, A., Usunier, N., Garcia-Duran, A., Weston, J., & Yakhnenko, O. (2013). Translating embeddings for modeling multi-relational data. In *NeurIPS 2013, Advances in neural information processing systems*. Curran Associates, Inc., https://proceedings.neurips.cc/paper_files/paper/2013/hash/1cecc7a77928ca8133fa24680a88d2f9-Abstract.html. [Accessed 06 December 2024].

El-Sappagh, S., Franda, F., Ali, F., & Kwak, K.-S. (2018). SNOMED CT standard ontology based on the ontology for general medical science. *BMC Medical Informatics and Decision Making*, *18*(1), 76. http://dx.doi.org/10.1186/s12911-018-0651-5.

Grover, Aditya, & Leskovec, Jure (2016). node2vec: Scalable feature learning for networks. CoRR abs/1607.00653. arXiv:1607.00653. http://arxiv.org/abs/1607.00653.

He, Bofan, Cheng, Jerry Q., & Gu, Huanying (2025). Static and dynamic embedding approaches to identify is-a relations in SNOMED CT. In *Will appear in the 13th IEEE international conference on healthcare informatics*. IEEE.

Ou, Mingdong, Cui, Peng, Pei, Jian, Zhang, Ziwei, & Zhu, Wenwu (2016). Asymmetric transitivity preserving graph embedding. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1105–1114). New York, NY, USA: Association for Computing Machinery, http://dx.doi.org/10.1145/2939672.2939751.

Perozzi, Bryan, Al-Rfou, Rami, & Skiena, Steven (2014). DeepWalk: Online learning of social representations. In *Proceedings of the 20th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 701–710). New York, NY, USA: Association for Computing Machinery, http://dx.doi.org/10.1145/2623330.2623732.

Tang, Jian, Qu, Meng, Wang, Mingzhe, Zhang, Ming, Yan, Jun, & Mei, Qiaozhu (2015). LINE: Large-scale information network embedding. In *Proceedings of the 24th international conference on world wide web* (pp. 1067–1077). Republic and Canton of Geneva, CHE: International World Wide Web Conferences Steering Committee, http://dx.doi.org/10.1145/2736277.2741093.

Tsitsulin, A., Mottin, D., Karras, P., & Müller, E. (2018). VERSE: Versatile graph embeddings from similarity measures. In *Proceedings of the 2018 world wide web conference on world wide web - WWW '18* (pp. 539–548). New York, NY, USA: ACM, http://dx.doi.org/10.1145/3178876.3186120.

Wu, H., Song, C., Ge, Y., & Ge, T. (2022). Link prediction on complex networks: An experimental survey. *Data Science and Engineering*, *7*(3), 253–278. http://dx.doi.org/10.1007/s41019-022-00188-2.